CSC 380 HW#4 Setting up the Development Environment

1. Change your Google CoLab subscription to "Pro, free 1-year subscription for students"

🎓 **How to Sign Up for Free Google Colab Pro (Students & Faculty)**

✅ **Eligibility**

- Must be a **student or faculty member** at a **U.S. college or university**.
- Must be able to **verify academic status**, typically using a .edu email address.

📝 **Sign-Up Steps**

1. **Visit the Colab for Higher Education Page**
   Go to [Google Colab for Higher Education](). [blog.google]
2. **Click the "No cost to students and educators" Button**
   This will take you to the verification process.
3. **Verify Your Academic Status**
   You'll be asked to confirm your identity using your **institutional email address** (e.g., ending in .edu).
4. **Activate Your Free Colab Pro Subscription**
   Once verified, you'll receive access to Colab Pro for **one full year**, including:
   - Faster GPUs
   - Longer runtimes
   - More memory
   - Slideshow Mode for live code presentations

2. Create an OpenAI account and put a small amount of money (e.g. $2) into the account.

Step 1: Create an OpenAI Account

1. Go to: 👉 https://platform.openai.com/signup
2. Sign up using:
   - Email + password OR
   - Google / Microsoft login
3. Verify:
   - Email address
   - Phone number (if prompted)

🔹 **Step 2: Log in to the OpenAI Platform**

After signing up, go to:
👉 https://platform.openai.com/
This is your main dashboard.

🔴 **Step 3: Fix the "insufficient_quota" Error (Critical)**

If you see this error:

RateLimitError: insufficient_quota (429)

👉 It means your account has **no available credits**

🔹 **Step 4: Add a Payment Method**

1. Go to: 👉 https://platform.openai.com/account/billing
2. Click: **"Add payment method"**
3. Enter:
   - Credit or debit card

🔹 **Step 5: Add Funds / Enable Usage**

After adding a card:

- Open: 👉 https://platform.openai.com/account/usage
- Then: 👉 https://platform.openai.com/account/limits

Set:

- **Soft limit:** $3
- **Hard limit:** $5

✅ This keeps your spending safe

💡 **Recommended: Add a Small Amount**

- Add **$2–$5**
- This is enough for many assignments

🔹 **Step 8: Wait and Retry**

- Wait ~1–2 minutes after adding billing
- Run your code again

✅ The error should disappear

3. Obtain API access keys to OpenAI, HuggingFace and Langchain

🔑 **How to Obtain API Access Keys**

1. 🧠 OpenAI API Key

Purpose: Access GPT models and embeddings.

Steps:

1. Go to **https://platform.openai.com/signup** and create an account.
2. After logging in, visit https://platform.openai.com/account/api-keys.
3. Click **"Create new secret key"**, name it, and copy the key.
4. You _***may***_ need to enter your **credit card** information, so you know.

2. 🤗 Hugging Face API Token

Purpose: Access models and datasets hosted on Hugging Face.

Steps:

1. Sign up at **https://huggingface.co/join**.

2. Go to https://huggingface.co/settings/tokens.
3. Click **“New token”**, choose **read** access, and copy the token.

3. ✅ **How to Get a LangChain API Key**

1. Go to **https://smith.langchain.com** and sign up.
2. After logging in, navigate to your **settings** and find the **API Keys** section.
3. Click **“Create API Key”**, name it, and copy the key.

4. Store API keys in CoLab “secret keys”

1. Open CoLab and click on the “secrets” button (a key-shaped) on the left.
2. Add a key/secret by specifying the name and your api key. One secret at a time.
3. Make sure the names are **“openai_api”**, **“HF_TOKEN”** and **“langchain_api”** (to go with the start-up code).